RESEARCH STATEMENT

Hanbaek Lyu

I work in discrete probability and in machine learning and AI, and I publish in the leading venues of both fields. My research comprises 49 published or accepted papers, with 7 more under review. In probability, my papers appear in top venues including the *Annals of Probability*, the *Annals of Applied Probability*, *Forum of Mathematics, Sigma*, *Transactions of the AMS*, and *IMRN*. In machine learning and AI, these include 18 papers in the field's top venues—8 at *ICML*, 3 at *NeurIPS*, 1 at *RLC*, 5 in *JMLR*, and 1 in *SIAM Journal on Optimization*—along with a *Nature Communications* paper featured as an Editor's Highlight. My research has been supported by **three NSF grants** (DMS-2010035, 2206296), most recently an **NSF CAREER Award** (DMS-2541878, $450,000, 2026–2031) on random matrices, Schrödinger bridges, optimal transport, and generative modeling, and also by an industry funding from KRAFTON AI on developing linear attention for long-context language models and Schrödinger-bridge-based generative modeling.

1. *Discrete Probability*: Emergence in Large-Scale Systems

In discrete probability, I establish scaling limits, phase transitions, and universality classes for large discrete systems. A recurring theme of my work is the discovery of unexpected connections between seemingly distant objects. Synchronization of coupled oscillators is connected to clustering in an annihilating particle system [ALS24], and its long-term behavior is governed by the persistence probabilities of a random walk with correlated increments [LS19]. Three-color excitable media can be viewed as a projection of Lamport's classical clock synchronization algorithm [GLS18]. A two-type diffusion-limited annihilating system can be analyzed through embedded token-passing and path-swapping processes [JJLS23]. The longest soliton lengths in randomized box-ball systems are connected to the extreme behavior of a circular exclusion process and to its diffusion limit, a semimartingale reflecting Brownian motion [LLP22, LLPS24]. And random matrices with given margins concentrate around static Schrödinger bridges, which can be computed by Sinkhorn's matrix scaling algorithm [LM25, DLP26b, DLP26a].

1.1. Random Matrices with Given Margins and Entropic Optimal Transport.

Large random matrices with prescribed row and column sums (margins) arise in contingency tables, network flows, and graphs with given degree sequences, and they exhibit phase transitions absent from classical random matrix models. In 2010, Barvinok conjectured that a small change in the margins of a random contingency table can trigger a sharp phase transition—a non-analytic change in its typical structure, suggestive of computational hardness. With Dittmer and Pak, I proved this conjecture in *Transactions of the AMS* [DLP20] (Figure 1); with Pak, I further showed that Good's 1950 independence heuristic for counting contingency tables fails dramatically beyond the transition [LP22].

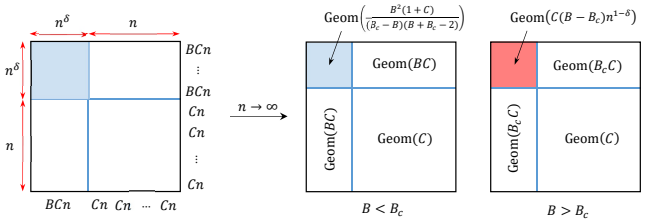


Figure 1. Phase transition in random contingency tables [DLP20]. (Left) A table whose first $\lfloor n^\delta \rfloor$ rows and columns have margins $\lfloor BCn \rfloor$ and whose remaining n rows and columns have margins $\lfloor Cn \rfloor$. (Right) Limiting distributions of the entries of the uniform table in the subcritical ($B < B_c$) and supercritical ($B > B_c$) regimes, for $1/2 < \delta < 1$; $\mathrm{Geom}(\lambda)$ is the geometric distribution with mean λ.

With Mukherjee [LM25], I then identified the principle underlying these phenomena. Our key innovation is a *transference principle*: a margin-conditioned random matrix is well approximated by a tractable matrix with independent entries, drawn from an exponential tilt of the base measure. Surprisingly, the required tilt is given by the two Schrödinger potentials of an implicit static Schrödinger bridge—an entropic optimal transport problem with suitable cost and divergence functions. This framework brings the tools of entropic optimal transport—Kantorovich duality, Schrödinger potentials, and Sinkhorn's algorithm—to previously intractable problems on random matrices with given margins. I regard this as one of my most significant contributions; it lays the foundation for the next five years of my research, supported by my NSF CAREER award (Section 3). The paper is under review in *Probability Theory and Related Fields*.

1.2. **Integrable Systems and Solitons.** Box-ball systems (BBS) are discrete integrable systems that originate from the Korteweg–de Vries (KdV) equation, crystallizing its classical soliton solutions in a combinatorial setting. I study how solitons emerge and evolve in such systems from a disordered random initial state. My work with Levine and Pike in *IMRN* [LLP22] is a foundational paper on randomized BBS, establishing the scaling limit of soliton lengths in the 1990 Takahashi–Satsuma BBS with i.i.d. initial configurations.

With Kuniba and Okado in *Nuclear Physics B* [KLO18], I extended this program to the most general BBS, associated with Kirillov–Reshetikhin crystals of type $A_n^{(1)}$, whose complete set of conserved quantities is an n-tuple of Young diagrams arising from rigged configurations, the action-angle variables of the system. For random initial states, we determined the limit shape of these Young diagrams by two independent routes—a Markov chain of carriers, and the thermodynamic Bethe ansatz applied to the fermionic formula—and showed that they agree: the limit shape is the logarithmic derivative of a deformed character of the Kirillov–Reshetikhin modules. With Kuniba in the *Journal of Statistical Physics* [KL20], I made this rigorous for the multicolor BBS by proving a large deviation principle for the row lengths, which gives almost sure convergence to an explicit Schur-polynomial limit shape at an exponential rate. The same argument explains why the two routes agree: the row lengths concentrate exponentially, while conditioning on highest states, as the Bethe ansatz requires, costs only a polynomial factor by a generalized ballot theorem.

With Lewis, Pylyavskyy, and Sen, in *Forum of Mathematics, Sigma* [LLPS24], I established the phase diagram and precise asymptotics of the top soliton lengths in the multicolor BBS. The analysis centers on the *multicolor carrier process*, a high-dimensional Markov chain in the integer orthant that behaves like a complex queueing system. By establishing its scaling limit and ruin probabilities, we obtained asymptotics for the soliton lengths down to the $\log\log n$ term, a precision beyond the reach of the thermodynamic Bethe ansatz. More recently, with Keating, Kim, and Loeser [KKLL25], I constructed a stochastic BBS that interpolates between the deterministic BBS, a discrete counterpart of the KdV equation, and PushTASEP, a particle system in the KPZ universality class, and proved its convergence to a semimartingale reflecting Brownian motion in the diffusive limit.

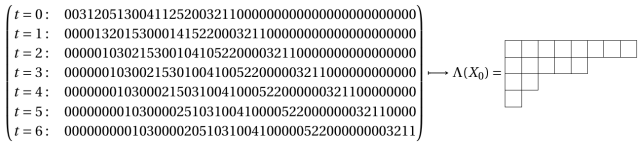


Figure 2. Evolution of a 5-color box-ball system, which decomposes into solitons of nondecreasing lengths. The soliton lengths are encoded by the invariant Young diagram on the right.

2. *Optimization and Machine Learning*: Algorithms to Theory to Applications

My research in optimization and machine learning centers on matrix factorization and its generalization to nonconvex constrained optimization. For these problems I develop principled algorithms with rigorous convergence guarantees, including online matrix and tensor factorization for Markovian data [LNB20, LSN22], supervised matrix factorization [LLY23, LLY24], stochastic first-order methods for correlated data [Lyu24, AL23, PL24], stochastic bilevel optimization [KKL24], and block majorization-minimization [LL25] with Riemannian extensions [LBNL26, LBNL24]. My work also bridges theory and practice by explaining why widely used algorithms work, including cyclic block optimization for nonconvex problems [LL25], policy optimization in reinforcement learning [PKXL25, PKXL26], coordinate-ascent maximum likelihood for phylogenetic inference [CJLRS26, CLR25], and Personalized PageRank for Koopman operator approximation [HLCL26].

2.1. **Cyclic Block Optimization.** Many large-scale optimization problems are solved by cyclic block coordinate methods, updating parameter blocks in a deterministic order. Despite their widespread empirical use, their theoretical convergence guarantees have long lagged behind those of randomized coordinate methods pioneered by Nesterov, Polyak, and Wright. With my student Li in the *SIAM Journal on Optimization* [LL25], I settled the convergence rate of cyclic block majorization-minimization for nonconvex, nonsmooth, constrained problems using a diminishing-radius framework. In *ICML* and *JMLR* [LBNL24, LBNL26], we extended this theory to Riemannian manifolds, establishing iteration complexity for both inexact first-order

Riemannian algorithms and Riemannian block majorization-minimization, and providing the first complexity bound for the Bures–JKO scheme in Wasserstein variational inference with non-strongly convex potentials.

2.2. **Stochastic Optimization.** Real-world data streams violate the i.i.d. assumptions on which classical stochastic optimization relies. In a series of *JMLR* papers [LNB20, LSN22, Lyu24], I developed a unified framework of stochastic block majorization-minimization (SBMM) with weakly convex and multi-convex surrogates, proving asymptotic stationarity and establishing convergence rates under Markovian and dependent data; this provides finite-sample guarantees for online matrix and tensor factorization, online expectation-maximization, and Nesterov's dual averaging.

Beyond Markovian mixing, I study stochastic methods under minimal data requirements. With Alacaoglu in *ICML* [AL23], I established convergence and complexity for projected stochastic subgradient methods for constrained nonconvex optimization driven by general dependent processes. With my student Powell in *ICML* [PL24], we showed that neither independence nor mixing is needed for optimal first-order rates—recurrence alone suffices: for regularized incremental surrogate optimization (MISO), we proved the optimal $O(n^{-1/2})$ rate under arbitrary recurrent sampling, with constants explicitly governed by the process's hitting and target times.

I also extend these stochastic techniques to hierarchical problems. With Jeongyeol Kwon and Dohyun Kwon in *ICML* [KKL24], I established the computational complexity of first-order methods in stochastic bilevel optimization, obtaining optimal sample complexity for single- and two-loop algorithms under relaxed smoothness conditions.

2.3. **Machine Learning Applications.** Building on my foundational work in discrete probability and nonconvex optimization, I develop machine learning methods with broad impact in network science, computational biology, and genomics. In network analysis, I introduced *network dictionary learning* (NDL) to uncover latent mesoscale structures by decomposing complex networks into a compact dictionary of interpretable motif patterns (Figure 3). The culmination of this work appeared in *Nature Communications* as an Editor's Highlight [LKVP24] (where a reviewer termed it a "tour de force"), combining MCMC sampling of subgraphs via random k-paths [LMS23] with online matrix factorization [LNB20]. In computational biology, we leveraged this framework to infer single-molecule long-range chromatin interactions from high-throughput genomic data [PPL+24]. Furthermore, using supervised matrix factorization developed in *NeurIPS* and *ICML* [LLY23, LLY24], I designed algorithms to identify latent gene clusters associated with patient clinical phenotypes, successfully discovering clinically known oncogenes.

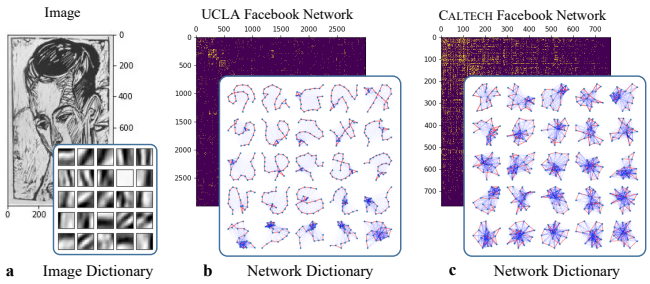


Figure 3. Dictionaries learned from square patches of an image (a) and from 20-node subgraphs of the UCLA (b) and CALTECH (c) Facebook networks [LKVP24].

3. Current and Future Research Program

My research over the next five years is organized around three directions. The first is the program of my NSF CAREER Award on random matrices, Schrödinger bridges, optimal transport, and generative modeling; the second carries my work on solitons and box-ball systems toward the central questions of integrable probability; the third is an industry collaboration with KRAFTON AI on the architecture of large language models.

3.1. **Random Matrices, Schrödinger Bridges, Entropic Optimal Transport, and Generative Modeling.** Many large-scale systems are observed only through marginal aggregates, from network traffic flows to unpaired cross-domain datasets in machine learning. One can infer the underlying joint distribution either at random (maximizing entropy subject to marginal constraints) or by design (minimizing an optimal transport cost). The central premise of my NSF CAREER program (DMS-2541878, 2026–2031) is that these two viewpoints converge: margin-constrained random matrices are governed by a hidden variational principle, the static Schrödinger bridge of entropic optimal transport. The program pursues a unified theory across three aims: (1) foundational theory and duality of generalized Schrödinger bridges with fast Sinkhorn-type algorithms; (2) transference principles from tractable tilted models to margin-constrained matrix ensembles; and (3) generative modeling via domain-adaptive Schrödinger bridges and random transport.

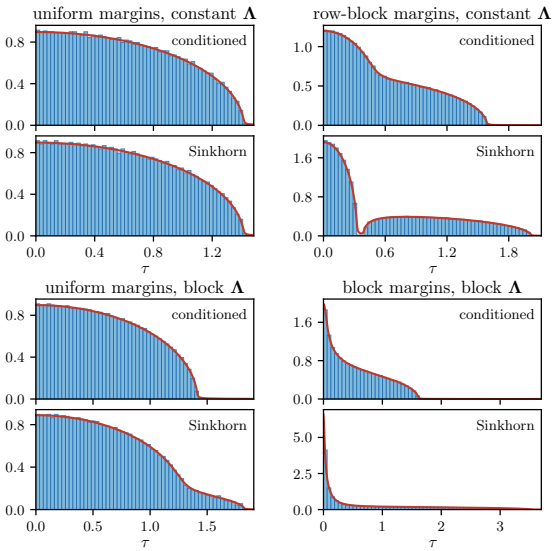


Figure 4. Singular values of the fluctuation matrix (histograms) and the density predicted by the matrix Dyson equation (red) for 5000×5000 tables, under four choices of margins and prior mean. In each pair, the upper panel shows a Poisson matrix conditioned on its margins [DLP26a], and the lower panel shows the same Poisson matrix rescaled to the same margins by Sinkhorn's algorithm [DLP26b]. The two models share a typical table but have different fluctuation spectra, both governed by the matrix Dyson equation.

Two recent papers with my students Duan and Powell establish the random-matrix foundation. For Sinkhorn-rescaled random matrices, we developed a quantitative stability theory for static Schrödinger bridges, proving bulk spectral rigidity and a central limit theorem for empirical Schrödinger potentials [DLP26b]. For margin-conditioned Poisson matrices, we established a transference principle to independent entries, a law of large numbers, and a functional central limit theorem [DLP26a]. Via Mallows-type random permutations, this resolves Pal's conjecture on limiting partition functions under weaker assumptions and settles Mukherjee's conjecture on the asymptotic normality of linear permutation statistics. While both ensembles share the same mean profile governed by entropic optimal transport, their fluctuation spectra differ, as the limiting variance profile for the matrix Dyson equation differ (Figure 4).

On the algorithmic side, with my student Bi and Jakwang Kim, I am developing stochastic Sinkhorn algorithms for entropic optimal transport with Lagrangian costs (such as graph shortest paths) where the full Gibbs kernel is intractable. A stochastic mesoscale approximation replaces the kernel by local averages over sampled blocks, reducing computational complexity while preserving accuracy via Schrödinger bridge stability [BKL26].

The third aim translates this theory to generative modeling. Dynamic Schrödinger bridges map between unpaired data distributions; in my NSF-supported 2025 summer REU, undergraduates demonstrated this through diffusion Schrödinger bridges for unpaired image translation (Figure 5). Moving forward, we are replacing the standard KL divergence with domain-adaptive divergences—for which Sinkhorn algorithms still achieve linear convergence [CL25]—and extending the framework to multi-marginal settings such as video generation, supported in part by industry funding from KRAFTON AI.



Figure 5. Unpaired `Dog→Human` translation using a dynamic Schrödinger bridge (2025 summer REU, NSF DMS-2206296).

3.2. **Box-Ball Systems and Integrable Probability.** Outside the CAREER program, my second major research thrust pushes my work on box-ball systems toward the central questions of integrable probability. For the deterministic box-ball system with random initial data, the fluctuation theory of solitons is now well developed: Olla, Sasada, and Suda showed that a tagged soliton converges to Brownian motion in the diffusive scale and satisfies a large deviation principle [OSS24]. My program investigates what happens when stochasticity is introduced directly into the dynamics (Figure 6). I am developing the fluctuation theory of such stochastic box-ball systems from two complementary perspectives: analytic invariance principles and algebraic Bethe ansatz methods. With Keating, Kim, and Loeser [KKLL25], I established a general invariance principle for d-dimensional Markov jump processes converging to semimartingale reflecting Brownian motions in convex polyhedra. It accommodates distinct boundary dynamics on each face intersection, resolving an overdetermined Skorokhod decomposition with more reflection vectors than faces. Applied to the stochastic box-ball system—a probabilistic interpolation between the Takahashi–Satsuma cellular automaton and PushTASEP—it yields a diffusive limit whose reflection matrix records a "solitonic bias" absent from PushTASEP, establishing a unified diffusion limit for particle systems with asymmetric collisions.

With Aguirre, Kim, and Paguyo [AKLP26], I examined the algebraic perspective. Box-ball systems with a stochastic carrier—which independently fails to pick up or drop off balls with prescribed probabilities—form a family of stochastic higher-spin vertex models, where the coordinate Bethe ansatz is the standard gateway to KPZ universality and exact fluctuation laws. Here, our investigation yielded a striking negative result: we completely classified all stochastic vertex weights for which the coordinate Bethe ansatz closes at every particle number, proving they are precisely the fusion weights of Corwin and Petrov. Crucially, this fusion subfamily stays a uniformly positive distance from the box-ball point at every carrier capacity. In particular, the ansatz does not close for the stochastic box-ball system at any capacity $J \geq 2$, the regime in which multi-particle solitons exist: plane-wave solvability breaks down strictly before the crystal limit, where Yang–Baxter integrability survives only in combinatorial form.

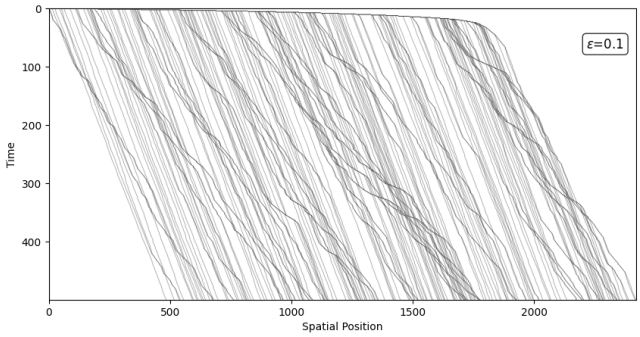


Figure 6. Ball trajectories in the stochastic box-ball system with infinite capacity and $\varepsilon = 0.1$, started from a block of 200 consecutive balls. The initial soliton breaks apart because the carrier occasionally fails to pick up balls; the long-term dynamics are governed by singleton balls performing independent random walks and by the spontaneous emergence of short solitons.

These results set the agenda for the next five years. While the invariance principle applies broadly, it does not exploit the hidden integrable structure of the box-ball system; conversely, the coordinate Bethe ansatz leverages that algebraic structure but does not survive carrier stochasticity. My aim is a fluctuation theory bridging the two, along three lines. First, I will extend the invariance principle to growing particle counts in the diffusive scaling, where the limit is an infinite-dimensional reflected diffusion, and determine whether these limits carry a random-matrix structure, as Warren's reflected interlacing Brownian motions do for the GUE corners process and, by the work of Gorin and Shkolnikov, for multilevel TASEP and the edge of multilevel Dyson Brownian motion [War07, GS15, GS14]. Second, the algebraic route remains open beyond plane-wave closure: our analysis in [AKLP26] produced exact formulas for these vertex models—including recursive generating functions at capacities $J = 2,3$ and closed forms for top Bethe amplitudes at every capacity—and I plan to build on them an alternative asymptotic analysis adapted to the near-box-ball corner of the parameter space. Third, because current carrier noise may disrupt combinatorial integrability too severely, I aim to identify milder stochastic perturbations that preserve soliton conservation laws, enabling fluctuation analysis with the combinatorial tools of the deterministic system.

3.3. **Machine Learning with KRAFTON AI.** Through an industry collaboration with KRAFTON AI, I investigate the architecture of large language models, translating insights from optimal transport, dynamical systems, and random matrices into practical deep learning architectures. The first project targets long-context language modeling via linear attention, which replaces quadratic softmax attention with a constant-memory recurrent state, eliminating the growing key–value cache at the cost of potential memory collisions. To resolve this trade-off, we introduced *Integral DeltaNet* (Figure 7), which pairs a fast delta-rule associative memory holding current key–value associations with a slow integral state that tracks historical memory trajectories over extended horizons. By dynamically combining both representations at readout, the model accesses information across multiple timescales while admitting a hardware-efficient, chunkwise-parallel formulation for training. Scaled to 340M and 1.3B parameters on 15B and 100B tokens, Integral DeltaNet consistently outperforms competitive baselines, including Kimi Delta Attention and Gated DeltaNet-2, in language modeling, commonsense reasoning, and retrieval benchmarks.

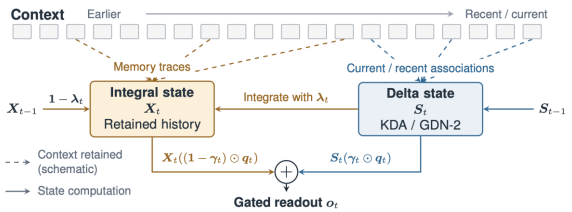


Figure 7. Integral DeltaNet. A fast delta state holds current key–value associations, and an integral state accumulates a weighted history of it; at readout, the model adaptively combines the two to access information across timescales.

The second project, with my student Moon, investigates *Sinkhorn attention* (or *Sinkformers*) [SABP22], replacing traditional one-sided softmax normalization with doubly stochastic Sinkhorn scaling so that each attention matrix forms a static Schrödinger bridge between uniform token distributions. Leveraging the quantitative potential stability theory developed in [DLP26b], we prove that finite-sample Sinkhorn attention converges with high probability to its population-level continuum limit, and we are currently extending these guarantees to unbounded token embeddings. The longer-term goal is a random-matrix foundation for attention mechanisms: establishing the spectral and fluctuation properties of Sinkhorn-rescaled kernel matrices, and deriving fundamental limits on memory capacity and expressivity in constant-state recurrent models.

These research directions provide accessible entry points across all academic levels: empirical generative modeling and linear attention benchmarks are well suited for undergraduate researchers, as demonstrated in my NSF-supported REU programs; algorithmic and discrete box-ball problems offer concrete milestones for doctoral students; and deep transference principles and random-matrix fluctuation theories provide rich programs for postdoctoral scholars and senior collaborators. Having supervised five doctoral students, two postdoctoral scholars, and twenty-four undergraduates across five REU cohorts, I cultivate a vertically integrated mentoring structure in which postdocs co-mentor graduate students and graduate students mentor undergraduate researchers.

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